

Your grade: 90%

Your latest: 90% • Your highest: 90% • To pass you need at least 80%. We keep your highest score.

Next item →

1. A Transformer Network processes sentences from left to right, one word at a time.

1 / 1 point

- ☒ False
- ☐ True

✓ Nice work

A Transformer Network can ingest entire sentences all at the same time.

2. Transformer Network methodology is taken from:

1 / 1 point

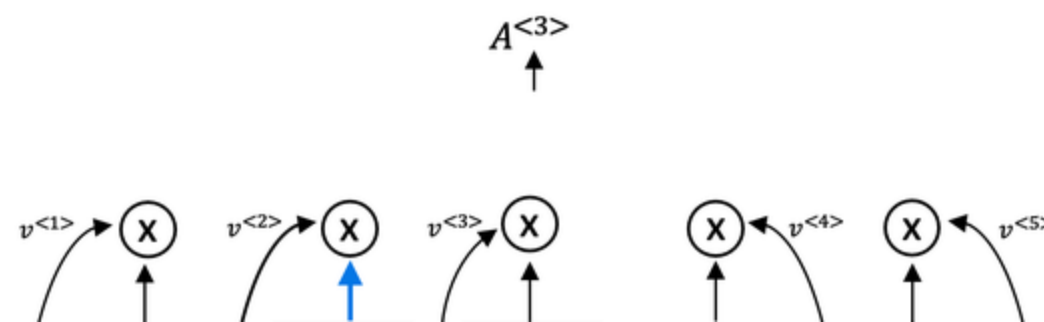
- ☐ GRUs and LSTMs
- ☐ RNN and LSTMs
- ☒ Attention Mechanism and CNN style of processing.
- ☐ Attention Mechanism and RNN style of processing.

✓ Nice work

Transformer architecture combines the use of attention based representations and a CNN convolutional neural network style of processing.

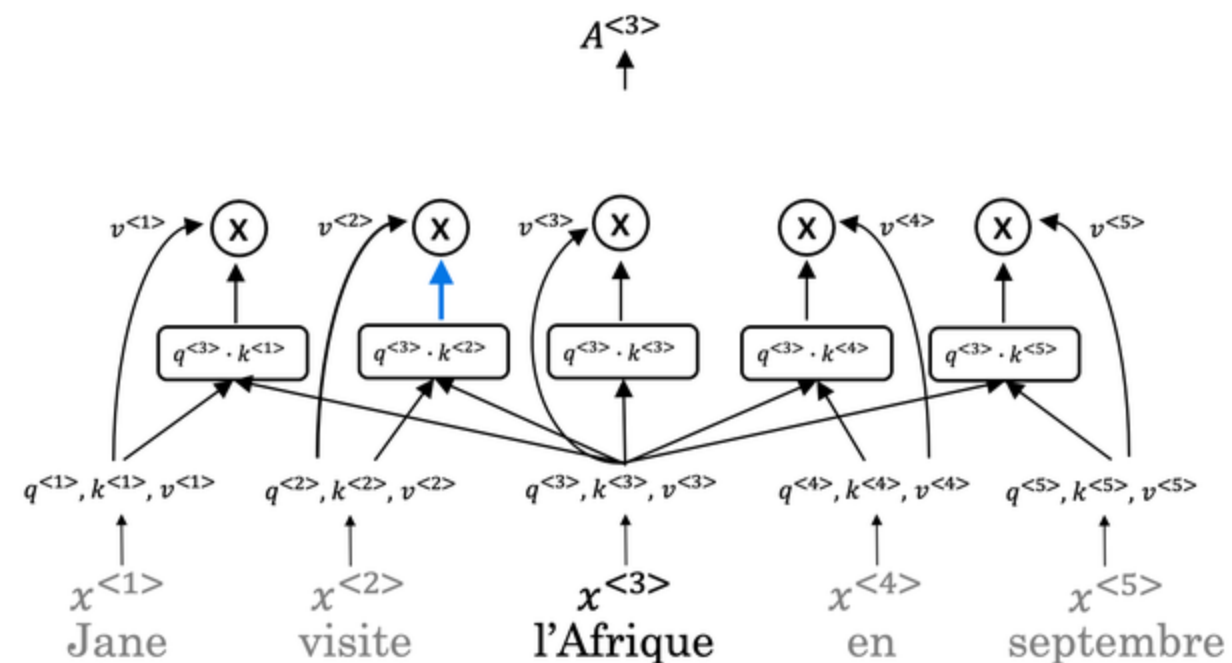
3. The concept of *Self-Attention* is that:

1 / 1 point



3. The concept of *Self-Attention* is that:

1 / 1 point



- ☐ Given a word, its neighbouring words are used to compute its context by taking the average of those word values to map the Attention related to that given word.
- ☐ Given a word, its neighbouring words are used to compute its context by selecting the lowest of those word values to map the Attention related to that given word.
- ☐ Given a word, its neighbouring words are used to compute its context by selecting the highest of those word values to map the Attention related to that given word.
- ☒ Given a word, its neighbouring words are used to compute its context by summing up the word values to map the Attention related to that given word.

✓ Nice work

4. Which of the following correctly represents *Attention* ?

1 / 1 point

- ☒ $Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$

4. Which of the following correctly represents *Attention* ?

1 / 1 point

- ☒ $Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$
- ☐ $Attention(Q, K, V) = softmax(\frac{QV^T}{\sqrt{d_k}})K$
- ☐ $Attention(Q, K, V) = min(\frac{QK^T}{\sqrt{d_k}})V$
- ☐ $Attention(Q, K, V) = min(\frac{QV^T}{\sqrt{d_k}})K$

✔ Nice work

5. Which of the following statements represents Key (K) as used in the self-attention calculation?

0 / 1 point

- ☒ K = interesting questions about the words in a sentence
- ☐ K = the order of the words in a sentence
- ☐ K = specific representations of words given a Q
- ☐ K = qualities of words given a Q

✘ Try again

To revise the concept watch the lecture *Self-Attention*; Q = interesting questions about the words in a sentence

$$Attention(W_i^Q Q, W_i^K K, W_i^V V)$$

1 / 1 point

6. What does i represent in this multi-head attention computation?

- ☐ The computed attention weight matrix associated with specific representations of words given a Q
- ☐ The computed attention weight matrix associated with the order of the words in a sentence
- ☐ The computed attention weight matrix associated with the i th “word” in a sentence.
- ☒ The computed attention weight matrix associated with the i th “head” (sequence)

6. What does i represent in this multi-head attention computation?

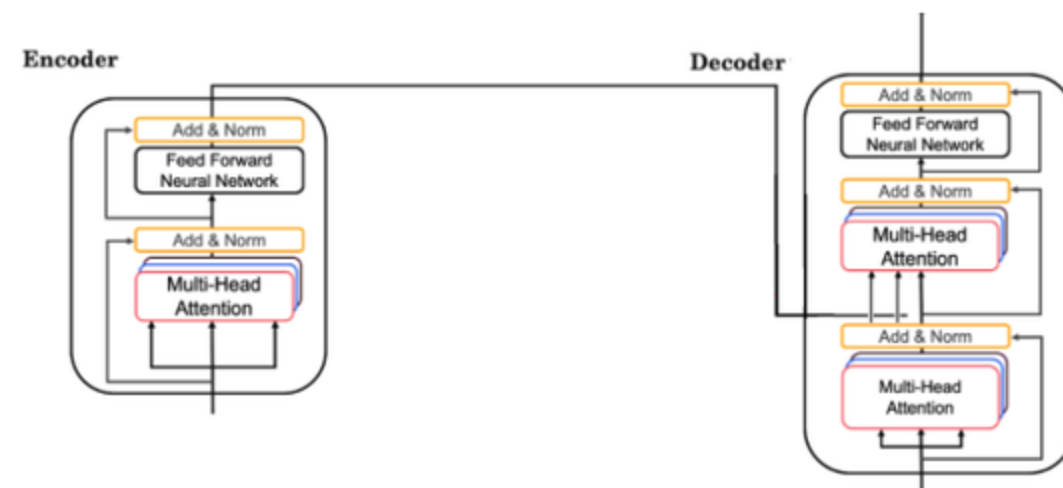
- ☐ The computed attention weight matrix associated with specific representations of words given a Q
- ☐ The computed attention weight matrix associated with the order of the words in a sentence
- ☐ The computed attention weight matrix associated with the i th “word” in a sentence.
- ☒ The computed attention weight matrix associated with the i th “head” (sequence)

✓ Nice work

i here represents the computed attention weight matrix associated with the i th “head” (sequence).

7. Following is the architecture within a Transformer Network (*without displaying positional encoding and output layers(s)*).

1 / 1 point



What is generated from the output of the *Decoder's* first block of *Multi-Head Attention*?

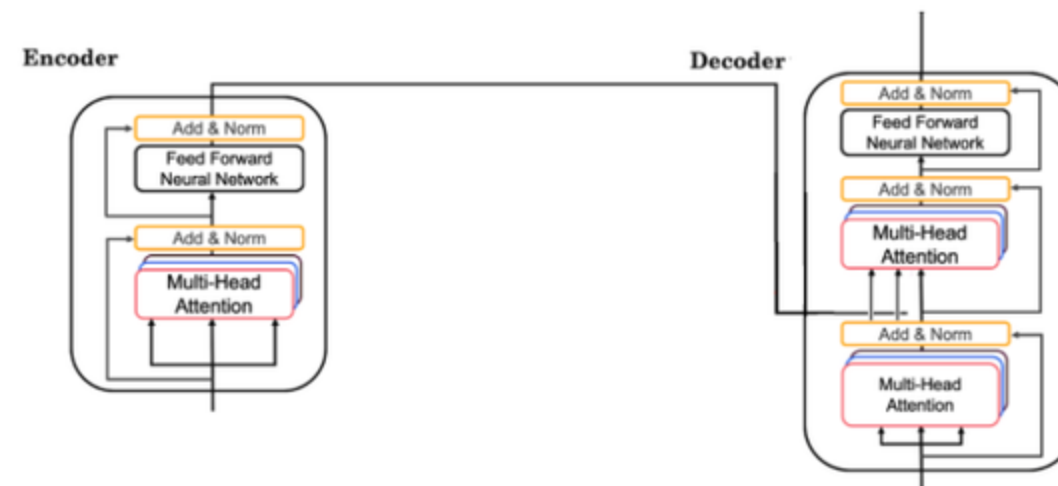
- ☒ Q
- ☐ K
- ☐ V

✓ Nice work

This first block's output is used to generate the Q matrix for the next Multi-Head Attention block.

8. Following is the architecture within a Transformer Network (*without displaying positional encoding and output layers(s)*).

1 / 1 point



What does the output of the *encoder* block contain?

- ☐ Softmax layer followed by a linear layer.
- ☐ Linear layer followed by a softmax layer.
- ☐ Prediction of the next word.
- ☒ Contextual semantic embedding and positional encoding information

✓ Nice work

The output of the *encoder* block contains contextual semantic embedding and positional encoding information.

9. Which of the following statements is true?

1 / 1 point

- ☐ The transformer network is similar to the attention model in that both contain positional encoding.
- ☒ The transformer network differs from the attention model in that only the transformer network contains

- ☒ Contextual semantic embedding and positional encoding information

✔ Nice work

The output of the *encoder* block contains contextual semantic embedding and positional encoding information.

9. Which of the following statements is true?

1 / 1 point

- ☐ The transformer network is similar to the attention model in that both contain positional encoding.
- ☒ The transformer network differs from the attention model in that only the transformer network contains positional encoding.
- ☐ The transformer network differs from the attention model in that only the attention model contains positional encoding.
- ☐ The transformer network is similar to the attention model in that neither contain positional encoding.

✔ Nice work

Positional encoding allows the transformer network to offer an additional benefit over the attention model.

10. Which of these is *not* a good criterion for a good positional encoding algorithm?

1 / 1 point

- ☐ Distance between any two time-steps should be consistent for all sentence lengths.
- ☒ It should output a common encoding for each time-step (word's position in a sentence).
- ☐ It must be deterministic.
- ☐ The algorithm should be able to generalize to longer sentences.

✔ Nice work

This is not a good criterion for a good positional encoding algorithm.